

Questions with factors of a number involved

1. Take a number and print its factors

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        System.out.println("Factors of " + n + " are:");
        for (int i = 1; i <= n; i++) {
            if (n % i == 0) {
                System.out.println(i);
            }
        }
    }
}
```

2. Take a number and find sum of its factors

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        int s=0;
        for (int i = 1; i <= n; i++) {
            if (n % i == 0) {
                s+=i;
            }
        }
        System.out.println(s);
    }
}
```

3. Take a number and count its factors

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        int c=0;
        for (int i = 1; i <= n; i++) {
            if (n % i == 0) {
                c++;
            }
        }
        System.out.println("Count of factors=" + c);
    }
}
```

```
}  
}
```

4. Take a number and check if it's prime number or not

```
import java.util.Scanner;  
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        System.out.print("Enter a number: ");  
        int n = sc.nextInt();  
        int c=0;  
        for (int i = 1; i <= n; i++) {  
            if (n % i == 0) {  
                c++;  
            }  
        }  
        if(c==2){  
            System.out.println("it's prime number");  
        }else{  
            System.out.println("Not Prime Number");  
        }  
    }  
}
```

5. Take a number and check if it's perfect number or not.

A number is said to be perfect if its sum of factors other than itself is equal to the number itself.

```
import java.util.Scanner;  
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        System.out.print("Enter a number: ");  
        int n = sc.nextInt();  
  
        int s=0;  
        for (int i = 1; i < n; i++) {  
            if (n % i == 0) {  
                s+=i;  
            }  
        }  
        if( s==n){  
            System.out.println("It's a perfect number");  
        }  
    }  
}
```

Sequences

Fibonacci: 0 1 1 2 3 5 8 upto nth term (every number is sum of previous two numbers)

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of terms: ");
        int n = sc.nextInt();

        int n1 = 0, n2 = 1, s;

        System.out.println("Fibonacci sequence:");
        System.out.print(n1 + " " + n2);
        for (int i = 1; i <= n; i++) {
            System.out.println(n1);
            s = n1 + n2;
            n1 = n2;
            n2 = s;
        }
    }
}
```

Pell Series: 0 1 1 5 12 29 upto nth term (Every number is a sum of second last and 2*last number)

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of terms: ");
        int n = sc.nextInt();

        int n1 = 0, n2 = 1, s;

        System.out.println("Pell series:");
        for (int i = 1; i <= n; i++) {
            System.out.println(n1);
            s = n1 + 2 * n2 ;
            n1 = n2;
            n2 = s;
        }
    }
}
```